

JOY SAHA

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Education

University At Albany, SUNY

PhD in Electrical and Computer Engineering

Major in Signal Processing and Communications

Advisor: Professor Daphney-Stavroula Zois

Aug 2024 – Present

NY, United States

CGPA: 4.00/4.00

Bangladesh University of Engineering and Technology

Bachelor of Science in Electrical and Electronic Engineering

Major in Communication and Signal Processing

Achieved an average CGPA of 3.75 across the last four terms.

Mar 2018 – May 2023

Dhaka, Bangladesh

CGPA: 3.61/4.00

Research Interests

- Feature Selection
- Statistical Signal Processing
- Optimization
- Machine Learning
- Sequential Decision Making
- Instant-wise Feature Grouping

Experience

Graduate Teaching Assistant

University at Albany, SUNY

Department of Electrical and Computer Engineering

Aug 2024 – Present

NY, United States

Research Project Assistant

The Research Foundation for SUNY

Department of Electrical and Computer Engineering

Summer 2025, 2026

NY, United States

Part-Time Lecturer

Presidency University

Department of Electrical and Electronic Engineering

Jan 2024 – Aug 2024

Dhaka, Bangladesh

Adjunct Lecturer

Bangladesh University of Textiles

Department of Textile Machinery Design and Maintenance

Feb 2024 – July 2024

Dhaka, Bangladesh

Projects

CSI-Based Sign Language Recognition Using CNN-GRU Architecture Enhanced with Attention | *WiFi CSI, Deep Sequential Models, Attention Mechanisms, Self-Supervised Learning* [GitHub](#)

- Designed a camera-free WiFi CSI-based sign language recognition system using CNN for spatial feature extraction across subcarriers and antennas, Bi-GRU with attention for temporal modeling, and self-supervised contrastive pretraining with extensive CSI augmentation, achieving up to 99% accuracy and validated through ablation studies and Grad-CAM interpretability.

Neural Networks: Investigating Effects of Layer Depth, Neuron Count, and Activation Functions | *Python, Feedforward Neural Networks, Backpropagation, Optimization* [GitHub](#)

- Implemented and evaluated feedforward neural networks from scratch, systematically analyzing the impact of layer depth, neuron count, and activation functions on classification performance, generalization, and computational efficiency across standard datasets to build intuition about data transformations through learned weights and activations.

Comparative Analysis of Ada-Hessian and First-Order Optimizers for CSI-Based Sign Language Recognition | *Second-Order Optimization, Ada-Hessian, Stochastic Estimation, Deep Learning* [GitHub](#)

- Conducted a comparative study of the Ada-Hessian second-order optimizer against first-order methods (SGD, Adam, AdamW, Adamax, Nadam, RMSprop) for WiFi CSI-based sign language recognition, demonstrating faster convergence (≈ 50 vs. $150+$ epochs), superior validation accuracy (99.31% on Lab dataset), and strong robustness to learning-rate variations despite higher per-iteration computational cost.

Social Distance Monitoring in COVID-19 Situation | *Matlab, CNN, YOLOv2, Image Processing* [GitHub](#)

- Developed a real-time social distance monitoring system using CNN and YOLOv2 to detect individuals in images/videos and visualize proximity compliance with color-coded bounding boxes.

Bayesian Hypothesis Testing on SeeClickFix Service Requests | *Bayes Criterion, Likelihood Ratio Test* [GitHub](#)

- Designed Bayes-optimal binary and 4-hypothesis tests to classify 529 citizen 311 service requests (Albany County SeeClickFix dataset) using multi-variate Bernoulli and multinomial likelihood models with Laplace-smoothed maximum-likelihood estimates, evaluated performance (P_F , P_D , P_M , P_e) across a grid of Bayes cost tuples, and traced ROC curves achieving AUC up to 0.995 with test errors as low as 1.9%.

IoT Based Environment Monitor System | *Arduino, NodeMCU, MQ135, DHT11, SIM800L, LCD* [GitHub](#)

- Built a solar-powered IoT system to monitor temperature, humidity, and air quality with cloud logging and GSM-based alert notifications upon threshold violations.

IoT Based Remote Heart Rate Monitoring System | *Arduino, ESP8266, AD8232* [GitHub](#)

- Implemented an IoT-enabled ECG monitoring system with on-device heart rate analysis, abnormal signal detection, and automated email-based data storage.

Undergraduate Thesis

Implementation-Friendly Convolution Neural Network for Sign Language Recognition

Using WiFi CSI Data | *Matlab, Python, CNN, Signal Processing* [GitHub](#) **April 2022 – May 2023**

- Developed a WiFi CSI-based sign language recognition system using signal processing and CNN models, leveraging MATLAB for data preprocessing and Python for model implementation.
- Achieved accuracy comparable to existing methods with reduced model complexity and simplified preprocessing, demonstrating an efficient and implementation-friendly approach.

Technical Skills

Languages: Python, C, C++, Matlab, Verilog

Simulation Tools: PSpice, TinkerCad, PSAF, PSSE, AutoCad, Proteus

Others: Word, Excel, PowerPoint, Latex, PLC, PCB Design

Publications

- Joy Saha**, Daphney–Stavroula Zois. “*Adaptive Sequential Feature Grouping and Acquisition for Cost-Efficient Classification.*” 2026 34th European Signal Processing Conference (EUSIPCO) (Accepted).
- Lamia Hossain, Ilma Hossain, Md. Shahidul Salim, S.M. Taslim Uddin Raju, **Joy Saha**. “*A Novel Technique for Classification of Motor Imagery EEG Signal Based on Deep Learning Approaches.*” International Conference on Big Data, IOT and Machine Learning (BIM 2023).

Relevant Coursework

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|----------------------------------|-----------------------------|--|
| • Probabilistic Machine Learning | • Optimization Methods and | • Machine Learning |
| • Deep Learning | Non-Linear Programming | • Probability and Random Processes |
| • Algorithms and Data Structures | • Digital Signal Processing | • Foundations of Statistical Inference |

Volunteer Work

Buet Rover Scout **Oct 2018**

Volunteer During Covid Pandemic **April 2020 – Aug 2020**

Organizer EEE Day **Feb 2023**

Awards and Scholarships

CNSE Excellence in Teaching **2025, 2026**

- Awarded Best Graduate Teaching Assistant, Department of Electrical and Computer Engineering, UAlbany.

Dean's List **Feb 2021 - April 2022**

- Acknowledged for achieving an average CGPA of 3.75 or above across consecutive terms(3-1, 3-2).

References

Dr. Daphney–Stavroula Zois

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Electrical & Computer Engineering
University at Albany, SUNY*

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📞 On Request

Dr. Hafiz Imtaiz

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Electrical & Electronic Engineering
Bangladesh University of Engineering and
Technology, Dhaka, Bangladesh*

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